

Guillain-Barré syndrome: Novel treatment by complement inhibition

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Abstract

Guillain-Barré syndrome (GBS) is an immune-mediated polyneuropathy that can cause acute flaccid tetraparesis and respiratory failure. Intravenous immunoglobulin and plasmapheresis were established as a standard of care, but 20% to 30% of patients require mechanical ventilation during the acute phase and up to 20% of patients cannot walk independently even 1 year after the onset. As the pathophysiology of GBS has been elucidated, complement activation draws attention as a key treatment point. In acute motor axonal neuropathy, anti-ganglioside antibody deposition and resulting complement activation lead to nerve conduction failure and eventually axonal degeneration. In acute inflammatory demyelinating polyneuropathy, although the process has not been clearly revealed, autopsied specimens showed that terminal complement complex rimmed the surface of the Schwann cells. Eculizumab is an antibody to the complement protein C5 and can potently inhibit complement activation. In a murine model with anti-ganglioside antibody-mediated neuropathy, eculizumab can avert respiratory failure and motor neuropathy. A randomized controlled phase II trial showed that eculizumab might be effective also for GBS patients. Inhibition of complement activation can be a new therapeutic option in the near future.

KEYWORDS

complement activation, eculizumab, gangliosides, Guillain-Barré syndrome

1 | INTRODUCTION

Guillain-Barré syndrome (GBS) is one of the most common causes of acute tetraparesis in the world.¹ It is an immune-mediated polyneuropathy typically preceded by infectious diseases.¹ GBS can affect all age groups.¹ During the acute phase, some patients must undergo mechanical ventilation due to respiratory failure leading to extended hospitalization.^{1,2} A large number of patients have long-term neurologic sequelae.² These facts suggest that GBS has certain impact on well-being and economy.

Intravenous immunoglobulin (IVIG) and plasmapheresis are still the current standard of care, which were established in the 1980s to 1990s.^{3,4} However, their mechanisms of action in treatment for GBS have not been clearly shown and the efficacy is not sufficient

especially in severe cases. Recently, numerous molecular targeted agents have been developed and put into practical use in various immune-mediated disorders. As the pathophysiology of GBS has been elucidated, a new therapeutic approach, which specifically targets major contributors to the pathophysiology, should be explored. This review focuses on complement activation in the pathophysiology of GBS, hoping that complement inhibition can be the novel therapeutic target.

2 | PROGNOSIS OF GBS

The disease course of GBS is monophasic.¹ Acute progressive weakness in the four limbs generally reaches its peak within 4 weeks from

the onset and shows trend toward progress even without immune modulatory treatment.¹ Thus, the prognosis of GBS is sometimes considered as positive. However, there are a considerable number of patients who have significant neurologic deficit² due to extensive damage of nerves during peak phase.

In fact, the proportions of the patients who cannot walk independently are 18% at 6 months and 16% at 12 months from the onset even after immune-modulatory treatments such as immunoglobulin and plasmapheresis.² These numbers might indicate that neurological improvement occurs mostly within the first 6 months and aftereffects last almost permanently after 1 year. In addition, almost 40% of patients need to change employment² and still experience pain at 1 year from the onset.⁵ These facts suggest that substantial economic loss can be caused by GBS.

Considering the aforementioned statements, we could conclude that the efficacy of the current treatment options for GBS is not sufficient and novel therapeutic interventions should be developed to reduce patients' neurologic impairment, facilitate their return to society, and improve their quality of life.

3 | HISTORY OF NEW TREATMENT DEVELOPMENT

A number of agents have been applied for treatment of GBS (Table 1). However, no drugs have proved to be effective for GBS.

In GBS, not only abnormalities in humoral-mediated immunity play an important role in the pathophysiology,^{1,6,7} but also cell-mediated immunity is presumed to be involved.⁸ Interferon (IFN)- β 1a was expected as a new therapeutic option based on the assumption

TABLE 1 Major therapeutic interventions developed for GBS and their status

Drug	Status
Past attempts	
Interferon β 1a	A small RCT could not show the efficacy ¹⁰
Brain-derived neurotrophic factor	A small RCT could not show the efficacy ¹¹
Mycophenolate mofetil	A small historical controlled clinical trial could not show the efficacy ¹³
Ongoing attempts	
Eculizumab	A small phase II RCT showed the possible efficacy. ²⁷ Further development is ongoing
Anti-C1q antibody	A phase Ib clinical trial is ongoing
IdeS	A phase II clinical trial is ongoing

Abbreviations: RCT, randomized clinical trial; IdeS, IgG-degrading enzyme of *S. pyogenes*.

of involvement of cellular immunity, increased tumor necrosis factor α , and decreased transforming growth factor β 1. Oral administration of IFN- β 1a in experimental allergic neuritis (EAN) shows a decrease in severity of neuropathy.⁹ However, significant differences in safety and efficacy between IFN- β 1a and placebo in the acute phase of GBS could not be distinguished by a small randomized clinical trial.¹⁰

Brain-derived neurotrophic factor (BDNF) was also applied to GBS, anticipating neuroprotective effects. A randomized controlled clinical trial was started.¹¹ However, during the trial, the manufacturer withdrew the drug; it failed in a clinical trial for motor neuron disease.¹² Because this trial for GBS registered a very limited number of cases, it is difficult to conclude whether BDNF is effective or harmful for GBS.¹²

Mycophenolate mofetil is an immunosuppressant medication that can suppress production of both activated T and B cells and is used to prevent acute transplant rejection in patients receiving allogeneic renal, hepatic, and cardiac transplants. A nonrandomized historical controlled clinical trial for GBS was conducted to study the efficacy and safety of mycophenolate mofetil.¹³ It compared 26 patients treated with mycophenolate mofetil and methylprednisolone (MP) and IVIG with 112 patients treated with MP and IVIG in a previous clinical trial;¹⁴ however, no significant differences were noted between the two groups as regards safety and efficacy outcomes.¹³

Additionally, there are case reports and several observational studies that reported treatments such as azathioprine,¹⁵ cyclophosphamide,^{16,17} and anti-T-cell antibody muromonab-CD3¹⁸ for GBS. However, none of these studies found the significant benefit or risk of these treatments.

4 | COMPLEMENT ACTIVATION IN GBS

Guillain-Barré syndrome is divided into two major subtypes: acute motor axonal neuropathy (AMAN) and acute inflammatory demyelinating neuropathy (AIDP).¹ The frequencies of each subtype vary significantly in the world. AMAN is frequently seen in Asian countries, such as China, Bangladesh, and Japan, whereas AIDP is common in European countries.⁶ Advances in understanding the immunopathogenesis and pathophysiology of AMAN have been made and its animal model has already been developed,¹⁹ whereas the pathophysiology of AIDP has not been fully understood.

A number of immunologic and bacteriologic studies revealed that molecular mimicry between GM1 and GD1a gangliosides on the axonal membrane and the lipooligosaccharides of *Campylobacter jejuni* plays a significant role in the pathophysiology of AMAN.⁶ Immune responses against *C. jejuni* infection induce autoantibodies which cross-react with gangliosides on the axolemma of the peripheral motor nerves. Those antibodies lead to complement activation and membrane attack complex deposition in the node of Ranvier on the axolemma. This immunologic cascade first causes functional block of nerve conduction based on disappearance of Na⁺ channel and detachment of paranodal myelin and later results in axonal degeneration.²⁰ Humanized monoclonal antibody eculizumab can specifically

bind to complement C5 to block its cleavage and potentially inhibit membrane attack complex formation.²¹ Eculizumab can effectively prevent respiratory paralysis and terminal motor neuropathy functionally and morphologically in anti-ganglioside GQ1b antibody-mediated neuropathy in a murine model.²²

In AIDP, the role of complement proteins and their activation has not been clearly shown. However, autopsy study showed complement activation and membrane attack complex deposition on Schwann cell membranes in patients with AIDP.²³ Moreover, demyelination and inflammation can be affected by complement suppression by cobra venom factor, which depletes C3 component, in EAN.^{24,25} Therefore, inhibition of complement activation can be effective also for AIDP.

5 | COMPLEMENT INHIBITION IN GBS PATIENTS

Based on the hypothesis that complement proteins can play a substantial role in disease progression of GBS, inhibition of complement activation can decrease severity during the acute phase. Especially for AMAN, formation of membrane attack complex in the axolemma can be the final step leading to axonal degeneration (Figure 1). Therefore, selective inhibition of membrane attack complex formation can be an efficient approach to avoid irreversible nerve damage, although there are several upstream mechanisms and cascades that have not been fully understood in the pathophysiology of AMAN.

Based on these, a phase II clinical trial was conducted to investigate the safety and efficacy of eculizumab, humanized monoclonal antibody to complement C5, for the acute phase of GBS in Japan (Table 1). It enrolled 34 patients with GBS who cannot walk independently within 14 days from the onset.^{26,27} Patients were randomly assigned (2:1) to IVIG plus eculizumab (900 mg) or placebo, which were administrated for 4 weeks.²⁶ The primary endpoint was efficacy, the proportion of patients who restored the ability of walking independently (functional grade ≤ 2) at week 4, and safety.²⁶

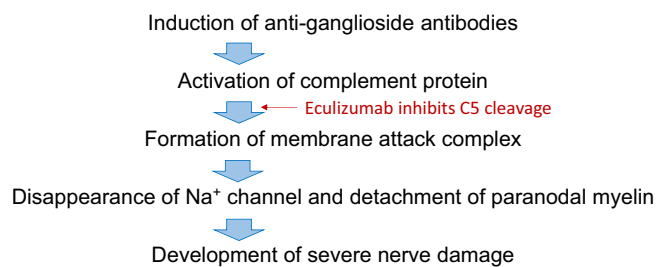


FIGURE 1 An immunologic cascade in acute motor axonal neuropathy (AMAN). There can be several pathways in the pathophysiology of AMAN. However, they can finally get to complement activation and resulting axonal degeneration. It can be an efficient approach to inhibit complement activation by eculizumab

The proportion of patients who can walk independently at week 4 was 61% in the eculizumab group and 45% in the placebo group. However, the result cannot be interpreted as statistically significant, because the primary efficacy endpoint could not exceed the predefined response rate.²⁷ On the other hand, the proportion of patients who could run at week 24 was significantly higher in the eculizumab (74%) than the placebo group (18%). For safety, the frequency of adverse events was not significantly different between the two groups. Brain abscess and anaphylaxis occurred as serious adverse effects in the eculizumab group and the causality could not be ruled out. Electrodiagnosis, such as AMAN or AIDP, did not correlate with the response to eculizumab. Therefore, eculizumab can be effective for severe cases with acute GBS regardless of AMAN or AIDP. At present, development of eculizumab for practical use in GBS is ongoing.

6 | FUTURE PERSPECTIVE

Eculizumab is a drug that is proven effective for GBS for the first time in 25 years. Inhibition of complement activation can be an efficient therapeutic strategy to treat GBS and so recently other complement inhibitors have been developed. A phase Ib, one-arm, open-label clinical trial of anti-C1q antibody is now enrolling patients in the United States (www.clinicaltrials.gov Identifier NCT04035135; Table 1). Furthermore, IgG-degrading enzyme of *Streptococcus pyogenes* (IdeS) has been also developed for GBS. IdeS is a cysteine protease that cleaves IgG antibodies into F(ab')₂ and Fc fragments.²⁸ In an in vitro study, IdeS cleaved IgG and blocked complement activation induced by anti-GM1, anti-GD1a, and anti-GQ1b IgG antibodies. A phase II, one-arm, open-label clinical trial of IdeS, imlifidase, is also enrolling patients in European countries (www.clinicaltrials.gov Identifier NCT01582763; Table 1).

At present, GBS is still considered a disease that can cause significant neurologic sequelae in a considerable number of patients. However, a study of eculizumab proved the dramatic improvement in the independent walking ability of GBS patients during acute phase. Additionally, several new drugs have been developed and therefore overcoming GBS in the future can be promising.

AUTHORSHIP DECLARATION

All authors are in agreement with the content of the manuscript.

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CONFLICT OF INTEREST

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